DHRUV CHAND

dhruvchand21@hotmail.com | (647)-985-5240 | linkedin.com/in/dhruv-chand1 | github.com/Dhruv-Chand

HIGHLIGHTS OF QUALIFICATIONS

- Graduated with Distinction from McMaster University's 4-year Honours Computer Science Co-op program with a **3.7 GPA**.
- 16 months of Software Engineering experience in **Evertz R&D** environment developing applications and automation tools.
- Skilled in **Python**, **C/C++**, **Java**, and **JavaScript**, with experience building both **front-end** and **back-end** systems.
- Knowledge in **TCP/IP Networking**, **Video Codecs**, **Media Containers**, **Colour Spaces**, and real-time streaming applications.
- Experience in **REST APIs**, **WebSockets**, Computer Vision, SQL, **Machine Learning**, and **Linux** systems.

EDUCATION

McMaster University, Hamilton ON — Sept 2021 - April 2026
B.A.Sc. Honours Computer Science Co-op — **Overall GPA: 3.7**
Relevant Course Work: Data Structures & Algorithms, Programming Fundamentals, Databases, Operating Systems, Computer Architecture, Computer Networks & Security, Machine Learning Applications, Compilers, and Information Security.

EXPERIENCE

Software Engineer Intern Co-op — May 2024 - August 2025
Evertz Microsystems — *16 Month Co-op*

- Developed **Python** and **Bash** automation scripts to safely reallocate and expand OS partitions on **Linux** based Evertz products, increasing available system space and reliability.
- Worked alongside **Application Engineers** to troubleshoot and resolve software, hardware, stream configuration, and networking issues on **ORT** devices in live broadcast environments (Overture Real Time devices).
- Applied strong **Linux** systems and networking fundamentals (processes, permissions, **TCP/IP**, multicast/unicast) to debug deployment, configuration, and runtime issues.
- Developed and maintained **React.js** interfaces using HTML, CSS, and **RESTful API's** for internal broadcast monitoring and diagnostic tools, improving usability and responsiveness across devices (based on PROGUARD System UI).

PROJECTS

RTSL | Real Time Sign Language Translator — September 2025 - April 2026
Python, Vue.js, OpenCV, PyTorch — *GitHub Repository*

- Developed a real-time ASL-to-English translation system leveraging **AI**, **Computer Vision**, and **Deep Learning** with **PyTorch**.
- Evaluated SVM, CNN, ResNet, and **Temporal Graph Convolutional Network (TGCN)** architectures, comparing live gesture-recognition performance to identify an accurate low-latency model achieving 80% validation accuracy.
- Engineered a **MediaPipe** based feature extraction pipeline, transforming hand landmarks into normalized keypoints, joint positions, and temporal motion vectors to train and feed a **TGCN** model.
- Built a **WebSocket** based streaming backend to transmit live video frames and return **AI Model Predictions** in real time.
- Selected as a **Highlight Project** at the Capstone Expo and featured in the official *McMaster Engineering Capstone Recap*.

OCR | Optical Character Recognizer — September 2025 - December 2025
Python, OpenCV, PyTorch — *GitHub Repository*

- Built an end-to-end **OCR** application that segments, preprocesses, and classifies printed characters from images across 62 classes.
- Designed and trained a **Convolutional Neural Network** (3 convolutional blocks with batch norm, ReLU, max-pooling, dropout) using PyTorch, improving character classification accuracy from 67% (HOG + feedforward baseline) to 90% test accuracy.
- Implemented a custom character-segmentation algorithm based on the Potential Segmentation Columns (PSC) method.
- Evaluated model accuracy, training loss, and misclassifications to identify recognition errors and guide model improvements.

Bacteriophage Model — November 2023
C++, PGPLOT

- Implemented an efficient version of the **Runge-Kutta algorithm** to approximate a system of Ordinary Differential Equations
- Highlighted the critical role of timing in introducing additional bacteria for the eradication of infected bacteria, emphasizing the significance of precise timing in Phage Therapy.

TECHNICAL SKILLS/STRENGTHS

Computer Languages: Python, JavaScript, C/C++, Java, HTML/CSS, Bash, SQL, Haskell, Prolog
Frameworks & Developer Tools: Git/GitHub, Linux, React.js, Windows, Office, Powershell, Node.js, Vim, Make, Gcc